IA Mathematics for Natural Sciences B:
Examples Sheet 5 - Infinite series, and Taylor series

Please send all comments and corrections to jmm232@cam.ac.uk.

Questions marked with a (∗) are difficult and should not be attempted at the expense of the other questions. A section marked with a (†) contains content that is unique to the Mathematics B course.

(†) Convergence of infinite series

1. (a) What does it mean to say that the infinite series $u_1 + u_2 + \cdots$ *converges*? What does it mean to say that the infinite series $u_1 + u_2 + \cdots$ *converges absolutely*? How are these two conditions related?

   (b) By directly evaluating the relevant partial sums, decide whether the following series:

$$\sum_{n=1}^{\infty} ar^n, \qquad \sum_{n=1}^{\infty} anr^n, \qquad \sum_{n=1}^{\infty} \left(\sqrt{n+1} - \sqrt{n}\right),$$

   where a, r are constant with $|r| < 1$, are (absolutely) convergent.

2. (a) State the *comparison test*, and use it to show the harmonic series diverges. Hence, show that the following definition of the Riemann zeta function:

$$\zeta(p) := \sum_{n=1}^{\infty} \frac{1}{n^p},$$

   with p real, converges if and only if $p > 1$. For what values of p is it absolutely convergent?

   (b) State, and prove, the *alternating series test* for series convergence. Hence, show that the following definition of the Dirichlet eta function:

$$\eta(p) := \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^p},$$

   with p real, converges if and only if $p > 0$. For what values of p is it absolutely convergent?

3. State the *ratio test*, and use it to show $\displaystyle\sum_{n=1}^{\infty} \frac{2^n}{n!}$ converges absolutely. To what does it converge?

4. (*Miscellaneous series*) Using an appropriate test in each case, determine whether the following series are (i) convergent; (ii) absolutely convergent:

   (a) $\displaystyle\sum_{n=1}^{\infty} \frac{n^2+1}{3n^2+4}$ (b) $\displaystyle\sum_{n=1}^{\infty} \frac{n^{10}}{2^n}$ (c) $\displaystyle\sum_{n=1}^{\infty} \frac{n^{10}}{n!}$

   (d) $\displaystyle\sum_{n=1}^{\infty} \frac{n!}{10^n}$ (e) $\displaystyle\sum_{n=1}^{\infty} \frac{5 \cdot 8 \cdot 11 \cdot \ldots \cdot (3n+2)}{1 \cdot 5 \cdot 9 \cdot \ldots \cdot (4n-3)}$ (f) $\displaystyle\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{\sqrt{n}}$

   (g) $\displaystyle\sum_{n=1}^{\infty} (-1)^{n+1} \left(\frac{2n+5}{3n+1}\right)^n$ (h) $\displaystyle\sum_{n=2}^{\infty} \frac{1}{n \log(n)}$ (i) $\displaystyle\sum_{n=2}^{\infty} \frac{1}{n^2 \log(n)}$

   (j) $\displaystyle\sum_{n=1}^{\infty} \frac{n^2+2n}{n^3+3n^2+1}$ (k) $\displaystyle\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{(2n-1)^2}$ (l) $\displaystyle\sum_{n=1}^{\infty} \left(\sqrt{n^4+1} - n^2\right)$

Taylor series

5. Clearly state Taylor's theorem, giving Lagrange's formula for the remainder term. Hence, obtain the first three non-zero terms in the Taylor series of $\log(x)$ about $x = 1$ by direct differentiation. Using this expansion, together with Lagrange's form of the remainder, show that:

$$|\log(3/2) - 5/12| \leq 1/64,$$

and hence give an approximation of $\log(3/2)$ valid to one decimal place.

6. Write down the Taylor series about $x = 0$ for the following functions, finding their range of convergence by appropriate tests in each case:

$$\text{(a) } e^x, \quad \text{(b) } \log(1+x), \quad \text{(c) } \sin(x), \quad \text{(d) } \cos(x), \quad \text{(e) } \sinh(x), \quad \text{(f) } \cosh(x), \quad \text{(g) } (1+x)^a,$$

where in the final part $a \in \mathbb{R}$ is any real number (ignore endpoints of the range of convergence in the final case, where convergence is subtle). What happens when a is a non-negative integer? Learn these series off by heart, and get your supervision partner to test you on them.

7. (*Miscellaneous Taylor series*) Using the most efficient method in each case (this rarely involves differentiating!), find the first three non-zero terms in the Taylor expansions of the following functions:

(a) $\dfrac{1}{\sqrt{1+x}}$ about $x = 0$; (b) $\dfrac{1}{(x^2+2)^{3/2}}$ about $x = 0$; (c) $x\sinh(x^2)$ about $x = 0$;

(d) $\tan(x)$ about $x = 0$; (e) $\log(\cos(x))$ about $x = 0$; (f) $(\cosh(x))^{-1/2}$ about $x = 0$;

(g) $\log(2+x)$ about $x = 0$; (h) $\sin^6(x)$ about $x = 0$; (i) $\arcsin(x)$ about $x = 0$;

(j) $\dfrac{\sinh(x+1)}{x+2}$ about $x = -1$; (k) $e^{\sin(x)}$ about $x = \pi/2$; (l) $\cosh(\log(x))$ about $x = 2$.

8. Find the value of the thirty-second derivative of $\cos(x^4)$ at $x = 0$.

9. Let $f(x)$ be a function which can be expanded as a Taylor series near $x = 0$. Find the first two non-zero terms in the Taylor series of $\log(1 + f(x))$ about $x = 0$, assuming $1 + f(0) > 0$, $f'(0) \neq 0$, and $f''(0)(1 + f(0)) \neq (f'(0))^2$. Why do we need these conditions?

10. Find the first three non-zero terms in a series approximation of $\log(1 + x + 2x^2) - \log(x^2)$, as $x \to \infty$.

11. (a) Using the Taylor series for $\log(1+x)$, show that $1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \frac{1}{6} + \frac{1}{7} - \cdots = \log(2)$.

    (b) $(*)$ Hence, by an appropriate sequence of transformations of the series in part (a), show that:

$$1 + \frac{1}{3} - \frac{1}{2} + \frac{1}{5} + \frac{1}{7} - \frac{1}{4} + \frac{1}{9} + \frac{1}{11} - \frac{1}{6} + \cdots = \frac{3}{2}\log(2).$$

    Comment on this result. [*Look up the Riemann rearrangement theorem!*]

12. $(*)$ Sketch the function $f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) = \exp(-1/x^2)$ for $x \neq 0$, and $f(0) = 0$. Show that this function is infinitely differentiable at $x = 0$, and find its Taylor series at $x = 0$. Comment on the general utility of Taylor series.

(†) Landau notation

13. Give the definition of Landau's big-O notation, '$f(x) = O(g(x))$ as $x \to x_0$'. Using the formal definition, decide which of the following statements are true.

    (a) $x = O(x^2)$ as $x \to 0$, (b) $x^2 = O(x)$ as $x \to 0$, (c) $x = O(x^2)$ as $x \to \infty$, (d) $x^2 = O(x)$ as $x \to \infty$.

14. Give the leading terms in an approximation to each of the following functions in the given limits, indicating the leading behaviour of the remainder in Landau's big-O notation:

$$\text{(a) } \frac{x^3 + x}{x + 2} \text{ as } x \to 0, \qquad \text{(b) } \frac{\cos(x) - 1}{x^3} \text{ as } x \to 0, \qquad \text{(c) } \frac{1 + 2x + 2x^2}{3x + 3} \text{ as } x \to \infty.$$

15. Show that $(x^3 + x^2 + 1)^{1/3} - (x^2 + x)^{1/2} = -\frac{1}{6}x + \frac{1}{72x} + O\left(\frac{1}{x^2}\right)$ as $x \to \infty$.

Newton-Raphson iteration

16. Give an explanation of the Newton-Raphson algorithm for root finding, including an appropriate sketch. Under what general conditions is it guaranteed that Newton-Raphson will converge to the root of interest? Prove that, when it converges to the root of interest, the Newton-Raphson method enjoys *quadratic convergence*.

17. (a) Find the value of the first iterate of Newton-Raphson iteration for the function $f(x) = x - 2 + \log(x)$ with a starting guess of $x_0 = 1$.

    (b) Find the value of the first and second iterates of Newton-Raphson iteration, valid to two decimal places, for the function $f(x) = x^2 - 2$ with a starting guess of $x_0 = 1$.

18. Let $f(x) = x^2 - 2\epsilon x - 1$, where ϵ is a small positive parameter ($0 < \epsilon \ll 1$). Let x_i be the ith Newton-Raphson iterate, with a starting guess of $x_0 = 1$, and let x_* be the unique positive root satisfying $f(x_*) = 0$. By Taylor expansion, show that $|x_i - x_*| \propto \epsilon^{n_i}$, where: (a) $n_0 = 1$; (b) $n_1 = 2$; (c) $n_2 = 4$.

19. ($*$) Consider the cubic equation $x^3 - 2x + 2 = 0$. Perform a numerical investigation (for example, by writing some simple code) to determine the ranges of starting guesses x_0 for Newton-Raphson which converge to the various roots, if any. Comment on the sensitivity of Newton-Raphson to the choice of initial guess x_0. [*Afterwards, look up Newton fractals - it is particularly interesting to see the behaviour of Newton-Raphson in the complex plane!*]

IA Mathematics for Natural Sciences B:
Examples Sheet 6 - Single-variable integration

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Riemann sums

1. Explain what is meant by a *Riemann sum* for a function $f : [a, b] \to \mathbb{R}$ using a *partition* $P = (x_0, ..., x_n)$ (with $x_0 = a, x_n = b$) and *tagging* $T = (t_1, ..., t_n)$. How can we use a sequence of Riemann sums to find the value of an integral from 'first principles'? Using appropriate sequences of Riemann sums in each case, find the integrals:

   (a) $\int_0^1 x\,dx$, (b) $\int_0^1 x^2\,dx$, (c) $\int_0^1 \sqrt{x}\,dx$, (d) $\int_0^1 \cos(x)\,dx$, (e) $\int_1^\infty \frac{dx}{x^{1+\alpha}}$, for $\alpha > 0$.

2. (∗) If a sequence of Riemann sums for a function $f : [a, b] \to \mathbb{R}$ converges, must the function f be integrable? Give a proof or counterexample.

The fundamental theorem(s) of calculus

3. By considering a small change in x, give a graphical proof of the *first fundamental theorem of calculus*:

$$\frac{d}{dx}\int_a^x f(t)\,dt = f(x),$$

   for constant a. Why does the right hand side not depend on a? Using the theorem, evaluate:

   (a) $\frac{d}{dx}\int_1^x \frac{\log(t)\sin^2(t)}{t^2+7}\,dt$, (b) $\frac{d}{dx}\left[\sum_{n=0}^{N}\binom{N}{n}\int_n^x \sin(y^2+y^6)\,dy\right]$, (c) $\frac{d}{dx}\left[\sin(x)\int_x^0 \sin(\cos(t))\,dt\right]$.

4. Without evaluating the integrals, determine the local extrema of the functions F_1, F_2 defined by:

   (a) $F_1(x) = \int_0^x t^2\sin^2(t)\,dt$, (b) $F_2(x) = \int_{-\infty}^x e^{-t^2}\,dt$.

   Hence, sketch the graphs of F_1, F_2. [*Assume that* $F_2(x) \to \sqrt{\pi}$ *as* $x \to \infty$*; we will prove this later!*]

5. Explain what is meant by an *indefinite integral* $F(x)$ of the function $f(x)$. Using the first fundamental theorem of calculus, deduce that if $F(x)$ is *any* indefinite integral of $f(x)$, we have:

$$\int_a^x f(t)\,dt = F(x) - F(a),$$

   which is the *second fundamental theorem of calculus*. Explain how we can use the theorem to find definite integrals without resorting to Riemann sums.

Standard integrals

6. Using your knowledge of derivatives, *write down* the indefinite integrals of each of the following functions, where $a \neq 0$, $\alpha \neq -1$ and f is any (differentiable, non-zero) function:

   (a) $(ax+b)^\alpha$ (b) e^{ax+b} (c) $(ax+b)^{-1}$ (d) $\sin(ax+b)$

   (e) $\cos(ax+b)$ (f) $\sec^2(ax+b)$ (g) $\operatorname{cosec}^2(ax+b)$ (h) $\sinh(ax+b)$

   (i) $\cosh(ax+b)$ (j) $f'(x)f(x)^\alpha$ (k) $\dfrac{f'(x)}{f(x)}$ (l) $\dfrac{1}{\sqrt{1-(ax+b)^2}}$

   (m) $\dfrac{1}{\sqrt{(ax+b)^2-1}}$ (n) $\dfrac{1}{\sqrt{1+(ax+b)^2}}$ (o) $\dfrac{1}{1+(ax+b)^2}$ (p) $\dfrac{1}{1-(ax+b)^2}$

   Learn all of these integrals off by heart, and get your supervision partner to test you on them.

7. Using the results of the previous question, evaluate the definite integrals:

   (a) $\displaystyle\int_0^2 (x-1)^2\,dx$ (b) $\displaystyle\int_0^\pi e^{i\theta}\,d\theta$ (c) $\displaystyle\int_0^\pi \cos(x)\,dx$

   (d) $\displaystyle\int_{-\pi/4}^{\pi/4} \sec^2(x)\,dx$ (e) $\displaystyle\int_0^1 \frac{2x+4}{x^2+4x+1}\,dx$ (f) $\displaystyle\int_0^1 \sinh(2x)\cosh^3(2x)\,dx$

   (g) $\displaystyle\int_0^\infty \frac{dx}{9+x^2}$ (h) $\displaystyle\int_0^1 \frac{dx}{\sqrt{x^2+x+1}}$ (i) $\displaystyle\int_0^3 \frac{8-2x}{\sqrt{6x-x^2}}\,dx$

8. By writing $\cos(bx)$ as the real part of a complex exponential, determine the indefinite integral of $e^{ax}\cos(bx)$. Similarly, find the indefinite integrals of $e^x(\sin(x)-\cos(x))$ and $e^x(\sin(x)+\cos(x))$.

9. (*Rational functions*) Explain the general strategy that one should adopt when integrating a rational function. Hence, determine the indefinite integrals of the following rational functions by decomposing into partial fractions:

   (a) $\dfrac{1}{1-x^2}$, (b) $\dfrac{3x}{2x^2+x-1}$, (c) $\dfrac{x^4+x^2+4x+6}{3+2x-2x^2-2x^3-x^4}$.

   Compare your answer to part (a) with your answer to Question 6(p). Are your results compatible?

Integration by substitution

10. Using substitutions, find: (a) $\displaystyle\int \frac{dx}{\sqrt{1-x^2}}$; (b) $\displaystyle\int \frac{dx}{\sqrt{1+9x^2}}$, verifying agreement with Question 6.

11. By means of appropriate substitutions in each case, or otherwise, determine the indefinite integrals of the following functions:

    (a) $x\sqrt{x+3}$, (b) $\tan(x)\sqrt{\sec(x)}$, (c) $\dfrac{e^x}{\sqrt{1-e^{2x}}}$, (d) $\dfrac{1}{x\sqrt{x^2-1}}$.

12. The *Weierstrass t-substitution* sets $t = \tan(\frac{1}{2}x)$. Show that under this substitution, $\sin(x) = 2t/(1+t^2)$, $\cos(x) = (1-t^2)/(1+t^2)$ and $dx/dt = 2/(1+t^2)$. Hence, find the indefinite integrals of:

    (a) $\operatorname{cosec}(x)$, (b) $\sec(x)$, (c) $\dfrac{1}{2+\cos(x)}$.

13. (*A subtle substitution*) For any sufficiently well-behaved function $f(x)$, prove the *Cauchy-Schlömlich formula*:

$$\int_{-\infty}^{\infty} f(x)\,dx = \int_{-\infty}^{\infty} f\left(x - \frac{1}{x}\right)\,dx,$$

by setting $u = x - 1/x$. [*Hint: sketch u as a function of x.*] Hence, show that $\displaystyle\int_{-\infty}^{\infty} \frac{x^2}{x^4+1}\,dx = \frac{\pi}{\sqrt{2}}$.

Integration by parts

14. State and prove the *integration by parts* formula. Using integration by parts, determine the following integrals:

$$\text{(a)} \int_{-\pi/2}^{\pi/2} x\sin(2x)\,dx, \qquad \text{(b)} \int_0^{\infty} xe^{-2x}\,dx, \qquad \text{(c)} \int_0^1 x\log\left(\frac{1}{x}\right)\,dx, \qquad \text{(d)} \int_0^{\infty} x^3e^{-x^2}\,dx.$$

15. By writing each of the following functions $f(x)$ in the form $1 \cdot f(x)$, and using integration by parts, determine their indefinite integrals:

(a) $\log(x)$, (b) $\log^3(x)$, (c) $\cosh^{-1}(x)$, (d) $\tanh^{-1}(x)$, (e) $\sin(\log(x))$.

Reduction formulae

16. (a) Show that for $n \geq 1$, we have: $\displaystyle\int \sin^n(x)\,dx = -\frac{1}{n}\cos(x)\sin^{n-1}(x) + \frac{n-1}{n}\int \sin^{n-2}(x)\,dx$. Hence, evaluate $\displaystyle\int \sin^6(x)\,dx$.

(b) Using part (a), show that $I_n = \displaystyle\int_0^{\pi/2} \sin^n(x)\,dx$ satisfies $I_n = (n-1)I_n/n$. Hence find I_2, I_4.

17. Establish reduction formulae for each of the following parametric integrals:

$$\text{(a) } I_n = \int_0^{\infty} x^n e^{-x^2}\,dx, \quad \text{(b) } J_n = \int_0^{\pi} x^{2n}\cos(x)\,dx, \quad \text{(c) } K_n = \int_0^{\infty} x^{n-1}e^{-x}\,dx, \quad \text{(d) } L_n = \int_0^{\infty} \frac{dx}{(1+x^2)^n}\,dx.$$

Using your reduction formulae, find I_3, I_5, J_3, J_5 and L_4. Establish also a general formula for K_n.

(†) The Leibniz integral rule

18. State Leibniz's integral rule for $\displaystyle\frac{d}{dx}\int_{a(x)}^{b(x)} f(x,t)\,dt$, and give a graphical proof. Verify the rule when:

(a) $a(x) = 0$, $b(x) = 1 + x$, and $f(x,t) = t(x-t)$;
(b) $a(x) = \pi x^2$, $b(x) = x$, and $f(x,t) = 2x^2t + x\sin(t)$.

19. Evaluate the limit: $\displaystyle\lim_{x\to\infty} \frac{d}{dx}\int_{\sin(1/x)}^{\sqrt{x}} \frac{2t^4+1}{(t-2)(t^2+3)}\,dt$.

20. (*Feynman's trick*) For all values of x, evaluate the integrals:

$$\text{(a) } f(x) = \int_0^1 \frac{t^x - 1}{\log(t)}\, dt, \qquad \text{(b) } g(x) = \int_0^\infty \frac{\log(1 + x^2 t^2)}{1 + t^2}\, dt, \qquad \text{(c) } h(x) = \int_0^1 \frac{\sin(x \log(t))}{\log(t)}\, dt.$$

(†) Integral inequalities

21. Using a sketch, show that $\sin(x) \geq 2x/\pi$ for $x \in [0, \frac{\pi}{2}]$. Hence prove: $\displaystyle\int_0^{\pi/2} \frac{x^2}{1 + \sin^2(x)}\, dx \leq \frac{\pi^2}{8}\left(1 - \frac{\pi}{4}\right)$.

22. State and prove *Schwarz's inequality* for integrals. Use it to show that: $\displaystyle\int_0^{\pi/2} \frac{\sin(x)}{\sqrt{x^2 + 1}} \leq \sqrt{\frac{\pi}{4} \arctan\left(\frac{\pi}{2}\right)}$.